1 Meeting #1 Notes

Exercises

Problem. Assume we are given the elliptic curve $E(\mathbb{K})$ in affine form and denote by $\overline{E}(\mathbb{K})$ the corresponding projective closure. What is $\#\overline{E}(\mathbb{K}) \setminus E(\mathbb{K})$?

Solution. We need to find such points on $\overline{E}(\mathbb{K})$ that are not included in $E(\mathbb{K})$. Recall that corresponding equations have the following form:

$$\begin{aligned} E(\mathbb{K}): &\quad y^2 = x^3 + ax + b, \\ \overline{E}(\mathbb{K}): &\quad Y^2 Z = X^3 + aXZ^2 + bZ^3, \end{aligned}$$

where $a, b \in \mathbb{K}$. Notice that given $(X : Y : Z) \in \overline{E}(\mathbb{K})$, we can almost always map it to $E(\mathbb{K})$ by using projection map $\pi : (X : Y : Z) \mapsto (X/Z, Y/Z)$. The only issue we encounter is when $Z = 0$: so $E(\mathbb{K})$ does not contain any points on $\overline{E}(\mathbb{K})$ with $Z = 0$. What are these points? Substitute $Z = 0$ into $\overline{E}(\mathbb{K})$: $X^3 = 0$, so $X = 0$. Note, however, that Y is arbitrary, so up to equivalence, we have the so-called *point at infinity* $(0 : 1 : 0)$. Thus we conclude $\#\left(\overline{E}(\mathbb{K}) \setminus E(\mathbb{K})\right) = 1$.

Problem. Prove that $\mathbb{F}_q^\times$ is cyclic (where $q = p^n$ generally for prime p).

Proof. For that, we prove the auxiliary lemma.

Lemma 1.1. Let H be a finite group of order n. Suppose that, for all divisors d of n, the set $\Omega_d = \{x \in H : x^d = 1\}$ has at most d elements. Then H is cyclic.

Proof. If there exists $x \in H$ of order d, the group $\langle x \rangle$ generated by x is cyclic of order d. In view of the hypothesis, all elements $y \in H$ such that $y^d = 1$ belong to $\langle x \rangle$. In particular, all elements of H of order d are generators of $\langle x \rangle$, and these are in number $\varphi(d)$. Hence, the number of elemets of H of order d is 0 or $\varphi(d)$. It cannot be 0 since otherwise $\sum_{d|n} \varphi(d)$ would show that the number of elements in H is less than n, contrary to the hypothesis. In particular, there exists an element $x \in H$ of order n and $H = \langle x \rangle$.

To prove our original theorem, simply use this lemma with $H = \mathbb{F}_q^\times$ and $n = q - 1$. Obviously, equation $x^d = 1$ has at most d solutions over $\mathbb{F}_q$.

Quadratic Reciprocity Law

The primary protocol where Quadratic Reciprocity Law is used is **Goldwasser-Micali cryptosystem**.

$\mathsf{KeyGen}(1^\lambda) \to (\mathsf{pk}, \mathsf{sk})$. One generates two large primes p and q, and computes $N = pq$. Then, one finds some non-residue x such that $\left(\frac{x}{p}\right) = \left(\frac{x}{q}\right) = -1$. Then, $\mathsf{pk} \leftarrow (x, N)$, while $\mathsf{sk} \leftarrow (p, q)$.

$\mathsf{Enc}(\mathsf{pk}, \mathbf{m} \in \{0,1\}^\ell) \to \mathbf{c} \in \mathbb{Z}_N^\ell$. Upon receiving a binary string $\mathbf{m}$ of length ℓ, one generates the ciphertext $\mathbf{c}$ as follows: $c_i \leftarrow y_i^2 x^{m_i}$ over $\mathbb{Z}_N$ where $y_i \leftarrow_R \mathbb{Z}_N^\times$.

$\mathsf{Dec}(\mathsf{sk}, \mathbf{c} \in \mathbb{Z}_N^\ell) \to \mathbf{m} \in \{0,1\}^\ell$. For each i, using the prime factorization (p, q), one determines whether the value c_i is a quadratic residue. If so, $m_i = 0$, otherwise $m_i = 1$.

Why this is secure? Given $x \in \mathbb{Z}_N$, there is no effective way to determine whether x is a quadratic non-residue, given no factorization of N.

Why given p, q we are good? Because x is a quadratic residue in $\mathbb{Z}_N$ iff it is both quadratic residue in $\mathbb{Z}_p$ and $\mathbb{Z}_q$. Thus, it suffices to check whether both $x^{(p-1)/2} = 1$ and $x^{(q-1)/2} = 1$.

2 Meeting #3 Notes

Problem. Assume $f(\underline{X}) \in \mathbb{Z}[X_1, \ldots, X_n]$ is a homogeneous polynomial in n variables of degree n. Let α be the coefficient of $(X_1 \ldots X_n)^{p-1}$ in $f(\underline{X})^{p-1}$. Prove that

$$\#\{\underline{X} \in \mathbb{F}_p^n : f(\underline{X}) = 0\} = (-1)^{n+1}\alpha \text{ in } \mathbb{F}_p$$

Consider the special case of elliptic curves.

Proof. Denote by $V := \{\underline{X} \in \mathbb{F}_p^n : f(\underline{X}) = 0\}$. Define $g(X) := 1 - X^{p-1}$. This function is obviously the characteristic function of $\{0\}$. In such case, we can count zero points simply by:

$$\#V = \sum_{\underline{X} \in \mathbb{F}_p^n} g \circ f(\underline{X}) = \sum_{\underline{X} \in \mathbb{F}_p^n} (1 - f(\underline{X})^{p-1}) = -\sum_{\underline{X} \in \mathbb{F}_p^n} f(\underline{X})^{p-1}$$

Let $f(\underline{X})^{p-1} = \sum_{\underline{u}} c_{\underline{u}} \underline{X}^{\underline{u}}$. Then, we have:

$$\begin{aligned}\#V &= -\sum_{\underline{X} \in \mathbb{F}_p^n} \sum_{\underline{u}} c_{\underline{u}} \underline{X}^{\underline{u}} = -\sum_{\underline{u}} \sum_{\underline{X} \in \mathbb{F}_p^n} c_{\underline{u}} \underline{X}^{\underline{u}} = -\sum_{\underline{u}} \sum_{\underline{X} \in \mathbb{F}_p^n} c_{\underline{u}} \prod_{j=1}^{n} X_j^{u_j} \\ &= -\sum_{\underline{u}} c_{\underline{u}} \prod_{j=1}^{n} \sum_{X_j \in \mathbb{F}_p} X_j^{u_j} = -\sum_{\underline{u}} c_{\underline{u}} \prod_{j=1}^{n} S(X^{u_j}),\end{aligned}$$

where $S(\cdot)$ is the function we used during Chevalley–Warning theorem proof. For the given $\underline{u}$, for product to be non-zero, every u_j must be divisible by $p-1$. Thus, the degree of the monomial must be at least $n(p-1)$. However, note that $\deg f(\underline{X})^{p-1} \leq (p-1)n$, so the only term that contributes is in fact $u_1 = u_2 = \cdots = u_n = p-1$. Therefore,

$$\#V = (-1)^{n+1} c_{p-1,\ldots,p-1} = (-1)^{n+1}\alpha$$

Application to Elliptic Curve. For Elliptic Curve, $f(X, Y, Z) = Y^2Z - X^3 - aXZ^2 - bZ^3$. In particular, $\#\{(X, Y, Z) \in \mathbb{F}_p^3 : f(X, Y, Z) = 0\} = \alpha$.

3 Meeting #4 Notes

These notes are based on [IR90, Section 1]. In this section, we define the unique factorization domain (UFD) and Euclidean domain, and show why $\mathbb{Z}$ and $k[x]$ are UFD and Euclidean domains.

Definition 3.1 (Euclidean domain)**.** We say that R is an **Euclidean domain** if there is a function $\lambda : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ such that if $a, b \in R$, $b \neq 0$, there exist $q, r \in R$ such that $a = qb + d$ and either $d = 0$ or $\lambda(d) < \lambda(b)$.

Further, we introduce the following notation: for some elements $a_1, \ldots, a_n \in R$ in ring R, let $\langle a_1, \ldots, a_n \rangle = \{\sum_{i=1}^{n} r_i a_i : r_i \in R\}$. We say I is finitely generated if there are $a_1, \ldots, a_n$ such that $I = \langle a_1, \ldots, a_n \rangle$. If $I = \langle a \rangle$ for some $a \in I$, we say that I is a principal ideal. We say that R is a principal ideal domain (PID) if every ideal $I \subseteq R$ is principal.

Theorem 3.2. Euclidean domain is PID.

Proof. Suppose R is an Euclidean domain and $I \subseteq R$ is an ideal. We shall prove that $I = \langle a \rangle$ for some $a \in I$. Let us construct it as follows: a is such that $\lambda(a)$ is the smallest in the set $\{\lambda(b) : b \in I \setminus \{0\}\}$. We prove that such choice suffices. First notice that $\langle a \rangle \subseteq I$ is due to the definition of ideal. To prove $I \subseteq \langle a \rangle$, we take arbitrary $b \in I$, $b \neq 0$. Divide b by a: there exist $q, r \in R$ such that $b = aq + r$ with $r = 0$ or $\lambda(r) < \lambda(a)$. Note that $r = b - aq \in I$, thus $\lambda(r) < \lambda(a)$ is not possible due to the choice of a. Thus, $r = 0$ and thus $b = aq \in \langle a \rangle$, so $I \subseteq \langle a \rangle$.

Further discussion now related to PID only. We introduce several new notions. *(i)* b divides a iff exists $c \in R$ such that $a = bc$ or $\langle b \rangle \subseteq \langle a \rangle$ *(ii)* $u \in R$ is a unit iff $u \mid 1$ or $\langle u \rangle = R$ *(iii)* $a, b \in R$ are associates if $a = bu$ for some unit $u \in R$ or $\langle a \rangle = \langle b \rangle$, *(iv)* $p \in R$ is irreducible if $a \mid p$ implies a is a unit or an associate of p, *(v)* $p \in R$ is prime if $p \neq 0$ and $p \mid ab$ implies $p \mid a$ or $p \mid b$.

Definition 3.3. d is said to be **greatest common divisor (GCD)** of two elements $a, b \in R$ if:

(a) $d \mid a$ and $d \mid b$.

(b) $d' \mid a$ and $d' \mid b$ implies $d' \mid d$.

Proposition 3.4 (Generalization of Bezout Identity)**.** Let R be a PID and $a, b \in R$. Then a and b have a greatest common divisor d and $\langle a, b \rangle = \langle d \rangle$.

Proof. Let us consider the ideal $\langle a, b \rangle$. Since R is a PID, there exists some d such that $\langle a, b \rangle = \langle d \rangle$. We prove that d is a GCD. First notice that $\langle a \rangle \subseteq \langle a, b \rangle = \langle d \rangle$ and similarly $\langle b \rangle \subseteq \langle d \rangle$, which translates to $d \mid a$ and $d \mid b$. Thus, the first condition for GCD is satisfied. Now suppose $d' \mid a$ and $d' \mid b$. Then, $\langle a \rangle, \langle b \rangle \subseteq \langle d' \rangle$. Thus, $\langle d \rangle = \langle a, b \rangle \subseteq \langle d' \rangle$, so $d' \mid d$. We are done.

Corollary 1. If $a, b \in R$ are relatively prime, $\langle a, b \rangle = R$.

Corollary 2. If R is a PID and $p \in R$ is irreducible, then p is prime.

Proof. Suppose $p \mid ab$ and p does not divide a (otherwise, we are done). Then, p and a are relatively prime, so by Corollary 1 one has $\langle a, p \rangle = R$. Thus, $\langle ab, pb \rangle = \langle b \rangle$. Since $ab \in \langle p \rangle$ and $pb \in \langle p \rangle$, we have $\langle b \rangle \subseteq \langle p \rangle$. Thus, $p \mid b$.

Now we show that every non-zero element of R is a product of irreducible elements. We do this in two steps: (i) if $a \in R$, $a \neq 0$, there is an irreducible dividing a, (ii) a is product of irreducibles.

Lemma 3.5. Let $\langle a_1\rangle \subseteq \langle a_2\rangle \subseteq \langle a_3\rangle \subseteq \dots$ be an ascending chain of ideals in PID R. Then the chain stabilizes in finitely many steps.

Proof. Let $I = \bigcup_{n\in\mathbb{N}}\langle a_n\rangle$. I is an ideal. Indeed, take any $b \in I$, then $b \in \langle a_k\rangle$ for some k. Then, trivially $\langle b\rangle \subseteq \langle a_k\rangle \subseteq I$. Since R is a PID, $I = \langle a\rangle$ for some $a \in R$. For the same argument as before, we get $I = \langle a\rangle \subseteq \langle a_k\rangle \subseteq I$ for some a_k. Thus, $I = \langle a_k\rangle$, and the result follows.

Proposition 3.6. Every non-zero element of R is a product of irreducibles.

Proof. Let $a \in R$, $a \neq 0$, a is not a unit. We first show that a is divisible by some irreducible element. If a is irreducible, we are done. Otherwise, $a = a_1b_1$ where a_1, b_1 are non-units. If a_1 is irreducible, we are done. Otherwise, $a_1 = a_2b_2$ where a_2, b_2 are non-units. If a_2 is irreducible, we are done. Otherwise continue as before. Notice that $\langle a\rangle \subseteq \langle a_1\rangle \subseteq \langle a_2\rangle \subseteq \dots$. By Lemma 3.5, this stabilizes after some a_k, which is irreducible.

We now show that a is the product of irreducibles. If a is irreducible, we are done. Otherwise let $p_1 \mid a$ with p_1 — irreducible. Then $a = p_1c_1$. If c_1 is a unit, we are done. Otherwise, let p_2 be an irreducible such that $p_2 \mid c_1$. Then, $a = p_1p_2c_2$. If c_2 is a unit, we are done. Otherwise, continue as before. Notice that $\langle a\rangle \subseteq \langle c_1\rangle \subseteq \langle c_2\rangle \subseteq \dots$. By Lemma 3.5, this stabilizes, so from some c_k, we have units. Since p_kc_k is irreducible, we are done.

Now we define the order function.

Definition 3.7. Define the order of a relatively to prime p (denoted as $\mathrm{ord}_p a$) as the largest integer n such that $p^n \mid a$, but not $p^{n+1} \mid a$.

Lemma 3.8. Order always exists.

Proof. Construct the converse. That is, for each $m > 0$, one has $a = b_mp^m$. Then, $pb_{m+1} = b_m$ so that $\langle b_1\rangle \subseteq \langle b_2\rangle \subseteq \langle b_3\rangle \subseteq \dots$ is an infinite ascending chain of ideals, which is impossible due to Lemma 3.5.

Lemma 3.9. If $a, b \in R$ with $a, b \neq 0$, then $\mathrm{ord}_p(ab) = \mathrm{ord}_p a + \mathrm{ord}_p b$.

Proof. Let $\alpha = \mathrm{ord}_p a$ and $\beta = \mathrm{ord}_p b$. Then, $a = p^\alpha c$ and $b = p^\beta d$ with p not dividing c, d. Note that $ab = p^{\alpha+\beta}cd$. Since p is prime, p does not divide cd.

Let S be a set of primes in R with the following two properties:

(a) Every prime in R is associate to a prime in S.

(b) No two primes in S are associate.

Example. In $\mathbb{Z}$, we choose S to be the set of positive primes. In S, we choose S to be the set of monic irreducible polynomials.

Property	$\mathbb{Z}$	$K[x]$	General Euclidean R
Unique factorization	$a = \pm \prod p^{e(p)}$	$f(x) = c \prod p(x)^{e(p)}$	$a = u \prod_{p \in S} p^{e(p)}$
Euclidean function λ	$\lvert \cdot \rvert$	$\deg(\cdot)$	–
Ideals	$n\mathbb{Z}$	$\langle f(x) \rangle$	Every ideal is principal
Divisibility	$a \mid b \iff b = ac$	$f \mid g \iff g = fh$	$a \mid b \iff \langle b \rangle \subseteq \langle a \rangle$
Units	$\{\pm 1\}$	$K^\times$	u s.t. $\langle u \rangle = R$
Associates	$a = \pm b$	$f \sim g \iff f = cg$	$a \sim b \iff \langle a \rangle = \langle b \rangle$
Irreducible elements	$2, 3, 5, 7, \ldots$	$x^2 + 1$ over $\mathbb{R}$	$a \mid p \implies \langle a \rangle = \langle p \rangle$ or R
GCD (Bezout)	$\gcd(a, b) = ua + vb$	$\gcd(f, g) = uf + vg$	d s.t. $\langle a, b \rangle = \langle d \rangle$
Canonical primes S	Positive primes	Monic irreducibles	Representative of associate class

Табл. 1: Comparison between $\mathbb{Z}$, $k[x]$, and a general integral domain/PID.

Theorem 3.10. Let R be a PID and S a set of primes with the properties given above. Then if $a \in R$, $a \neq 0$, we can write

$$a = u \prod_{p \in S} p^{e(p)},$$

where u is a unit. The unit u and the exponents $e(p)$ are uniquely determined by a. In fact, $e(p) = \mathrm{ord}_p a$.

Proof. Existence follows immediately from aforementioned discussion.
Take $\mathrm{ord}_q(\cdot)$ to both sides with $q \in S$. Then,

$$\mathrm{ord}_q a = \mathrm{ord}_q u + \sum_{p \in S} e(p) \mathrm{ord}_q p$$

Now, $\mathrm{ord}_q(u) = 0$ from the definition, and $\mathrm{ord}_q p = 1$ iff $q = p$. Thus, $\mathrm{ord}_q a = e(q)$. Since the exponents $e(q)$ are uniquely determined so is the unit u. This completes the proof.

3.1 Other Euclidean Domains

Theorem 3.11. Let $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ be the ring of Gaussian integers. Then, it is an Euclidean domain for $\lambda(z) = z\overline{z}$.

Proof. For $\alpha = a + bi$ and $\gamma = c + di$, we have $\alpha/\gamma = r + si$ for some rational r, s. Choose integers $m, n \in \mathbb{Z}$ such that $|r - m| \leq \frac{1}{2}$ and $|s - n| \leq \frac{1}{2}$ (simply by rounding). Set $\delta := m + ni$. Clearly, $\delta \in \mathbb{Z}[i]$. Moreover, $\lambda(\frac{\alpha}{\gamma} - \delta) = (r - m)^2 + (s - n)^2 \leq \frac{1}{2}$. Set $\rho := \alpha - \gamma\delta$. Then, $\rho \in \mathbb{Z}[i]$ and either $\rho = 0$ or $\lambda(\rho) = \lambda(\gamma(\frac{\alpha}{\gamma} - \delta)) = \lambda(\gamma)\lambda(\frac{\alpha}{\gamma} - \delta) \leq \frac{1}{2}\lambda(\gamma) < \lambda(\gamma)$.

Theorem 3.12. Let $\mathbb{Z}[\omega] = \{a + b\omega : \omega^2 + \omega + 1 = 0\}$ be the ring of Eisenstein integers. Then, $\mathbb{Z}[\omega]$ is an Euclidean domain with $\lambda(z) = z\overline{z}$.

4 Meeting #6 Notes

4.1 Algebraic Numbers and Integers

Definition 4.1. An **algebraic number** is a complex number α which is a root of some polynomial with rational coefficients $f \in \mathbb{Q}[x], f \neq 0$. **Algebraic integer** is a complex number ω which is a root of some polynomial with integer coefficients $g \in \mathbb{Z}[x], g \neq 0$.

Proposition 4.2. A rational number $r \in \mathbb{Q}$ is an algebraic integer iff $r \in \mathbb{Z}$.

Proof. Clearly integer $r \in \mathbb{Z}$ is an algebraic integer (since it is a root of $x - r$). Now suppose $r \in \mathbb{Q}$ is an algebraic integer, thus it is a root of some $\sum_{j=0}^{n} a_j x^j$, thus $\sum_{j=0}^{n} a_j r^j = 0$ with $a_j \in \mathbb{Z}$. Since r is rational, assume $r = \frac{p}{q}$. Thus $q^n \sum_{j=0}^{n} a_j \frac{p^j}{q^j} = 0$ or $\sum_{j=0}^{n} a_j q^{n-j} p^j = 0$. Note that q divides all terms but last, so $q \mid p^j$. Since $\gcd(p, q) = 1$, this implies $q \mid p$, thus $r \in \mathbb{Z}$.

Definition 4.3. A subset $V \subseteq \mathbb{C}$ of the complex numbers is called a $\mathbb{Q}$-module if it is a finite dimensional space over $\mathbb{Q}$. Denote by $[\gamma_1, \gamma_2, \ldots, \gamma_\ell]$ the set of all expressions $\sum_{i=1}^{\ell} r_i \gamma_i$ with $r_i \in \mathbb{Q}$ and $\gamma_i \in \mathbb{C}$.

Proposition 4.4. Let $V = [\gamma_1, \ldots, \gamma_\ell]$ and suppose that $\alpha \in \mathbb{C}$ has the property that $\alpha\gamma \in V$ for all $\gamma \in V$. Then, α is an algebraic number.

Proof. Since $\alpha\gamma_i \in V$, we have $\alpha\gamma_i = \sum_{j=1}^{\ell} a_{i,j}\gamma_j$ or simply $\alpha\boldsymbol{\gamma} = A\boldsymbol{\gamma}$, which implies $\det(A - \alpha E_{\ell\times\ell}) = 0$. Since $A \in \mathbb{Q}^{\ell\times\ell}$, α is a root of some polynomial with rational coefficients, and thus an algebraic number.

Proposition 4.5. The set of algebraic numbers forms a field.

Proof. Suppose α_1 and α_2 are algebraic numbers. We shall show that $\alpha_1 + \alpha_2$ and $\alpha_1\alpha_2$ are algebraic numbers as well. Suppose $\alpha_1^n + r_{n-1}\alpha_1^{n-1} + \cdots + r_0 = 0$ and $\alpha_2^m + s_{m-1}\alpha_2^{m-1} + \cdots + s_0 = 0$, where all $r_i, s_i \in \mathbb{Q}$. Let $V := [\alpha_1^i \alpha_2^j]_{(i,j)\in[n]\times[m]}$ be the $\mathbb{Q}$-module. For $\gamma \in V$ let us show that $\alpha_1\gamma \in V$ and $\alpha_2\gamma \in V$. Indeed, suppose $\gamma = \sum_{i,j} c_{i,j}\alpha_1^i\alpha_2^j$, then $\alpha_1\gamma = \sum_{i,j} c_{i,j}\alpha_1^{i+1}\alpha_2^j$. It would be clearly in V if we did not have α_1^n term, which is not included in V by definition. However, $\alpha_1^n \in V$ since $\alpha_1^n = -\sum_{j\in[n]} r_j\alpha_1^j$, thus $c_{n,j}\alpha_1^n\alpha_2^j \in V$, concluding the argument.

Therefore, $(\alpha_1 + \alpha_2)\gamma \in V$ and $(\alpha_1\alpha_2)\gamma \in V$. By proposition above, this implies that both $\alpha_1 + \alpha_2$ and $\alpha_1\alpha_2$ are algebraic numbers.

Finally, if α_1 is an algebraic number, α_1^{-1} is algebraic number as well. Indeed, $1 + r_{n-1}\alpha_1^{-1} + \cdots + r_0\alpha_1^{-n} = 0$.

Note that proof for algebraic integers is the same, but we have a ring structure.

Let Ω be the ring of algebraic integers. If $\omega_1, \omega_2, \gamma \in \Omega$, we say that $\omega_1 \equiv \omega_2 \pmod{\gamma}$ if $\omega_1 - \omega_2 = \gamma\alpha$ for some $\alpha \in \Omega$.

Proposition 4.6. If $\omega_1, \omega_2 \in \Omega$ and $p \in \mathbb{Z}$ is prime, then

$$(\omega_1 + \omega_2)^p \equiv \omega_1^p + \omega_2^p \pmod{p}$$

Proof. $(\omega_1+\omega_2)^p = \sum_{j=0}^{p} \binom{p}{j} \omega_1^j \omega_2^{p-j}$. Again, p divides all terms but the first and last, so we reach the conclusion.

Proposition 4.7. If α is an algebraic number then α is the root of a unique monic irreducible $f \in \mathbb{Q}[x]$. Furthermore, if $g \in \mathbb{Q}[x]$, $g(\alpha) = 0$, then $f \mid g$.

Proof. Let f be any monic irreducible with $f(\alpha) = 0$ (it exists because α is the root of some $q \in \mathbb{Q}[x]$, and if q is not irreducible, we can decompose it into factors where α will also be a root). Let us prove the second assertion first. If $f \mid g$ does not hold, then $\gcd(f, g) = 1$, and so $f(x)h(x) + g(x)t(x) = 1$ for some $h, t \in \mathbb{Q}[x]$. Setting $x = \alpha$ gives the contradiction. Uniqueness follows immediately.

The polynomial defined in proposition is called the **minimal polynomial** of α. α is called the algebraic number of degree n if corresponding f has a degree n. Now, define the field:

$$\mathbb{Q}(\alpha) = \{g(\alpha)/h(\alpha) : g, h \in \mathbb{Q}[x], h(\alpha) \neq 0\}$$

Proposition 4.8. If $\alpha \in \Omega$, then $\mathbb{Q}(\alpha) \cong \mathbb{Q}[\alpha]$.

Proof. Clearly $\mathbb{Q}[\alpha] \subseteq \mathbb{Q}(\alpha)$. If $h(\alpha) \in \mathbb{Q}[\alpha]$, $h(\alpha) \neq 0$, then $f(x)$ does not divide $h(x)$ where $f(x)$ is the minimal polynomial of α. Then, $\gcd(h(x), f(x)) = 1$ and so $s(x)f(x)+t(x)h(x) = 1$ for some $s, t \in \mathbb{Q}[x]$. Set $\alpha = 1$, thus $t(\alpha)h(\alpha) = 1$, so $h(\alpha) \in \mathbb{Q}[\alpha]^\times$. If $\beta \in \mathbb{Q}(\alpha)$, then $\beta = g(\alpha)h(\alpha)^{-1}$ and thus $\beta \in \mathbb{Q}[\alpha]$ since $h(\alpha)$ is a unit.

Corollary. If α is of degree n, then $[\mathbb{Q}(\alpha) : \mathbb{Q}] = n$.

4.2 The Quadratic Character of 2

We shall prove the following fact differently.

Proposition 4.9. $(2/p) = (-1)^\varepsilon$ where $\varepsilon \equiv (p^2-1)/8 \pmod 2$.

Proof. Let $\zeta = e^{2\pi i/8}$ and let $\tau = \zeta + \zeta^{-1}$. Clearly, both ζ and τ are algebraic integers, so we might work with congruences in the ring of algebraic integers. Let p be the odd prime in $\mathbb{Z}$ and notice that

$$\tau^{p-1} = (\tau^2)^{(p-1)/2} = 2^{(p-1)/2} \equiv (2/p) \pmod p$$

It thus follows that $\tau^p \equiv (2/p)\tau \pmod p$. Here, we used the fact that $\tau^2 = (\zeta+\zeta^{-1})^2 = 2$, and the last congruence is due to Euler's criterion. By earlier proposition, $\tau^p \equiv \zeta^p + \zeta^{-p} \pmod p$. But since $\zeta^8 = 1$, we have $\zeta^p+\zeta^{-p} = \zeta+\zeta^{-1}$ for $p \equiv \pm 1 \pmod 8$ and $\zeta^p+\zeta^{-p} = \zeta^3+\zeta^{-3} = -(\zeta+\zeta^{-1})$ for $p \equiv \pm 3 \pmod 8$. Summarizing,

$$\zeta^p + \zeta^{-p} = \begin{cases} \tau, & \text{if } p \equiv \pm 1 \pmod 8, \\ -\tau, & \text{if } p \equiv \pm 3 \pmod 8. \end{cases}$$

Thus, substituting into $\tau^p \equiv (2/p)\tau \pmod p$ we get $(-1)^\varepsilon \tau \equiv (2/p)\tau \pmod p$ with $\varepsilon \equiv \frac{p^2-8}{1} \pmod 3$. Multiplying both sides by τ we get $(-1)^\varepsilon 2 \equiv (2/p)2 \pmod p$, but dividing both sides by 2 we finally get $(2/p) = (-1)^\varepsilon$, which is a desired result.

4.3 Quadratic Gauss Sums

We shall generalize the relation $(\zeta + \zeta^{-1})^2 = 2$ for arbitrary odd prime p. Further, denote by $\zeta = e^{2\pi i/p}$ a primitive pth root of unity (root of $x^p + 1 = 0$).

Lemma 4.10. $\sum_{t=0}^{p-1} \zeta^{at}$ equals p if $a \equiv 0 \pmod{p}$. Otherwise, it is 0.

Proof. If $a \equiv 0 \pmod{p}$, then $\zeta^a = 1$, so the result follows. If $a \not\equiv 0 \pmod{p}$, then $\zeta^a \neq 1$, and so we get the geometric series sum $\sum_{t=0}^{p-1} \zeta^{at} = \frac{\zeta^{ap}-1}{\zeta^a-1} = 0$.

Corollary. Denote $\delta(x, y)$ the function which is 1 iff $x \equiv y \pmod{p}$. Then,

$$\delta(x, y) = p^{-1} \sum_{t=0}^{p-1} \zeta^{t(x-y)}$$

Lemma 4.11. $\sum_t (t/p) = 0$ where (t/p) is the Legendre symbol.

Proof. By definition, $(0/p) = 0$. Of the remaining $p - 1$ terms, the half are $+1$, the half is -1, so the result follows.

Definition 4.12. $g_a = \sum_t (t/p)\zeta^{at}$ is called the *quadratic Gauss sum.*

Proposition 4.13. $g_a = (a/p)g_1$.

Proof. If $a \equiv 0 \pmod{p}$, then $\zeta^{at} = 1$, and so $g_a = \sum_t (t/p) = 0$ by Lemma above. Suppose in contrary that $a \not\equiv 0 \pmod{p}$. Then,

$$\left(\frac{a}{p}\right) g_a = \sum_t \left(\frac{at}{p}\right) \zeta^{at} = \sum_x \left(\frac{x}{p}\right) \zeta^x = g_1.$$

Here we used the fact that $x \mapsto ax$ is a bijection. But since $(a/p)^2 = 1$, the result follows by multiplying both sides by (a/p).

Further, denote g_1 by g. If follows from proposition above that $g^2 = g_a^2$ for all $a \not\equiv 0 \pmod{p}$.

Proposition 4.14. $g^2 = (-1)^{(p-1)/2} p$.

Proof. The idea of the proof is to evaluate $\sum_a g_a g_{-a}$ in two ways.

If $a \not\equiv 0 \pmod{p}$, then $g_a g_{-a} = (a/p)(-a/p)g^2 = (-1/p)g^2$. Thus,

$$\sum_a g_a g_{-a} = \left(\frac{-1}{p}\right)(p-1)g^2$$

On the other hand,

$$g_a g_{-a} = \sum_x \sum_y \left(\frac{x}{p}\right)\left(\frac{y}{p}\right) \zeta^{a(x-y)}$$

Apply $\sum_a$ to both sides:

$$\sum_a g_a g_{-a} = \sum_x \sum_y \left(\frac{x}{p}\right)\left(\frac{y}{p}\right) \sum_a \zeta^{a(x-y)} = \sum_x \sum_y \left(\frac{x}{p}\right)\left(\frac{y}{p}\right) \delta(x, y)p = (p-1)p$$

Thus we arrive at the result.

Let $p^* = (-1)^{(p-1)/2}p$. The equation $g^2 = p^*$ is the desired analog of the equation $\tau^2 = 2$. Let $q \neq p$. We shall prove the following theorem.

Theorem 4.15. $(p/q)(q/p) = (-1)^{(p-1)(q-1)/4}$.

Proof. Let $p^* = (-1)^{(p-1)/2}$, as above. Now note:

$$g^{q-1} = (g^2)^{(q-1)/2} = (p^*)^{(q-1)/2} \equiv (p^*/q) \pmod{q}$$

Thus, $g^q \equiv (p^*/q)g \pmod{q}$. Now note:

$$g^q = \left(\sum_p (t/p)\zeta^t\right)^q = \sum_p (t/p)^q \zeta^{qt} \equiv g_q \pmod{q}$$

Thus it follows that $g^q \equiv g_q \equiv (q/p)g \pmod{q}$ and so

$$(q/p)g = (p^*/q)g \pmod{q}$$

Multiply both sides by g and use $g^2 = p^*$:

$$(q/p)p^* = (p^*/q)p^* \pmod{q},$$

implying that $(q/p) = (p^*/q) \pmod{q}$ or simply $(q/p) = (p^*/q)$. This proves the theorem. To see this, note that:

$$(p^*/q) = (-1/q)^{(p-1)/2}(p/q) = (-1)^{(q-1)(p-1)/4}(p/q)$$

5 Meeting #8 Notes

5.1 Zeta Function

Definition 5.1. Let V be a non-singular, projective variety over $K = \mathbb{F}_q$. Let N_s be the set of points rational over $K_s = \mathbb{F}_{q^s}$. The **(Hasse-Weil) zeta function** of V is the series given by:

$$Z(V,T) = \exp\left(\sum_{s>1} \frac{N_s T^s}{s}\right)$$

In particular, [IR90] considers $f \in \mathbb{F}_q[T_0, \ldots, T_n]$ — a homogeneous polynomial, and considers projective hypersurface $V := \#\overline{H}_f(\mathbb{F}_{q^s}) \subseteq \mathbb{P}^n(\mathbb{F}_{q^s})$.

5.2 Examples

Example (Single Point Variety)**.** Let V be a one-element set. Then, $Z(T) = 1/(1-T)$.

Example. Let $V = \mathbb{A}^d_K$, then $N_s = q^{sd}$, and thus $Z_V(T) = 1/(1-q^dT)$.

Example (Hyperplane)**.** Consider the hyperplane at infinity:

$$\overline{H}_f(\mathbb{F}_{q^s}) := \{(x_0, \ldots, x_n) \in \mathbb{P}^n(\mathbb{F}_{q^s}) : x_0 = 0\}.$$

The number of points is $N_s = q^{s(n-1)} + q^{s(n-2)} + \cdots + q^s + 1$. It follows that:

$$\sum_{s=1}^{\infty} \frac{N_s u^s}{s} = \sum_{m=0}^{n-1} \sum_{s=1}^{\infty} \frac{(q^m u)^s}{s} = -\sum_{m=0}^{n-1} \log(1 - q^m u),$$

where we used the identity $\sum_{s=1}^{\infty} w^s/s = -\log(1-w)$. Thus,

$$Z_f(u) = \prod_{m=0}^{n-1} \frac{1}{1 - q^m u}$$

Example. Consider the hypersurface $-y_0^2 + y_1^2 + y_2^2 + y_3^2 = 0$. It can be computed that the number of points on such surface is $N_s = q^{2s} + q^s + 1 + \chi_s(-1)q^s$. Thus:

$$\sum_{s=1}^{\infty} \frac{N_s u^s}{s} = \sum_{s=1}^{\infty} \frac{(q^s u)^s}{s} + 2\sum_{s=1}^{\infty} \frac{(qu)^s}{s} + \sum_{s=1}^{\infty} \frac{u^s}{s}$$

If -1 is a square in $\mathbb{F}_q$, then $\chi_s(-1) = 1$ for all s, otherwise $\chi_s(-1) = -1$ for s odd and $\chi_s(-1) = 1$ for all s even. In the first case, $Z(u) = (1-q^2u)^{-1}(1-qu)^{-2}(1-u)^{-1}$, while in the second $Z(u) = (1-q^2u)^{-1}(1-qu)^{-1}(1+qu)^{-1}(1-u)^{-1}$.

Example. For elliptic curve E over $K = \mathbb{F}_q$, there are algebraic integers $\alpha, \beta \in \mathbb{C}$ such that:

$$Z(E/K, T) = \frac{(1-\alpha T)(1-\beta T)}{(1-T)(1-qT)}$$

with $|\alpha| = |\beta| = \sqrt{q}$.

Lemma 5.2 (Hasse's Bound)**.** For any elliptic curve $E(\mathbb{F}_q)$, one has:

$$|\#E(\mathbb{F}_q) - 1 + q| \leq 2\sqrt{q}$$

Proof. In general, note:

$$\frac{TZ'(T)}{Z(T)} = \sum_{s>0} N_s T^s,$$

so $Z(V,T)$ actually encodes the number of points on V. Now, for $E(\mathbb{F}_q)$, we know $Z(V,T)$ explicitly, so let us find its logarithm:

$$\log Z(V,T) = \log(1-\alpha T) + \log(1-\beta T) - \log(1-T) - \log(1-qT)$$

Find formal derivative of both sides:

$$\frac{Z'(T)}{Z(T)} = \frac{1}{1-T} + \frac{q}{1-qT} - \frac{\alpha}{1-\alpha T} - \frac{\beta}{1-\beta T}$$

Expand $\frac{1}{1-T} = \sum_{n\geq 0} T^n$, $\frac{q}{1-qT} = \sum_{n\geq 0} q^{n+1}T^n, \ldots$ and use the fact that $Z'(T)/Z(T) = \sum_{s>0} N_s T^{s-1}$. We obtain

$$\#E(\mathbb{F}_{q^k}) = q^k + 1 - (\alpha^k + \beta^k)$$

Using that $|\alpha^k + \beta^k| \leq 2q^{k/2}$, we obtain:

$$|\#E(\mathbb{F}_{q^k}) - (q^k + 1)| \leq 2\sqrt{q^k}$$

5.3 Historical Notes

Recall the fact that $\sum_{n>0} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$.

Definition 5.3. Let A be an algebra of finite type over $\mathbb{Z}$. This means that $A \cong \mathbb{Z}[T_1, \ldots, T_m]/I$ for $I \subseteq \mathbb{Z}[T_1, \ldots, T_m]$. Define:

$$\zeta_A(s) = \prod_{\mathfrak{m} \subseteq A} \frac{1}{1 - \#(A/\mathfrak{m})^{-s}}$$

Example. For $A = \mathbb{Z}$, we have a standard Riehmann function.
By taking $X := \operatorname{Spec}(A)$, we get that $\mathfrak{m} = \{x\}$ are closed points and $A/\mathfrak{m}$ is a finite field of some order q_x. Then,

$$\zeta_X(s) = \prod_{x \in \operatorname{Spec}(A)} \frac{1}{1 - q_x^{-s}}$$

5.4 Conjectures

The Weil (F.K. Schmidt 1931) was able to prove that for $d := \deg f$:

$$Z_f(T) = \frac{P(T)}{(1-T)(1-qT)}, \quad P \in \mathbb{Z}[T],\ \deg P = (d-1)(d-2)$$

If C is a curve, $\deg P = 2g$, $g = \operatorname{genus}(C)$.

Proposition 5.4 (11.1.1 [IR90]). The zeta function is rational iff there exist complex numbers α_i and β_j such that $N_s = \sum_j \beta_j^s - \sum_i \alpha_i^s$.

Proof. On one hand $TZ'(T)/Z(T) = \sum_{s>0} N_s T^s$, on the other substitute $Z(T) = \frac{\prod_i (1-\alpha_i T)}{\prod_j (1-\beta_j T)}$.

Conjectures. Let V be a non-singular, projective variety of dimension d defined over the $K = \mathbb{F}_q$.

1. *Rationality.* Zeta function is a rational function with coefficients in $\mathbb{Z}$ in $T := q^{-s}$:
$$Z(V/K, T) = \frac{\prod_{j=1}^{d} P_{2j-1}(T)}{\prod_{j=1}^{d} P_{2j}(T)}, \quad P_j \in \mathbb{Z}[T],\ P_0 = 1 - T,\ P_{2d} = 1 - q^d T.$$
2. *Riemann hypothesis.* The polynomials $P_1, \ldots, P_{2d-1}$ factor over $\mathbb{C}$ as $P_k(T) = \prod_j (1 - \alpha_{k,j} T)$ with $|\alpha_{k,j}| = \sqrt{q^k}$.
3. *Functional equation.* Zeta function should satisfy:
$$Z(V, n-s) = \pm q^{nE/2 - Es} Z(V, s),$$
where E is the Euler characteristic of V.
4. *Betti numbers.* Degrees of P_i are determined by cohomological characterization of V.

5.5 Hasse-Davenport Relation

Theorem 5.5. Let $K := \mathbb{F}_q$, $K_s := \mathbb{F}_{q^s}$. Let $\chi' = \chi \circ N_{K_s/K}$ be the multiplicative character of K_s, while $\psi' = \psi \circ \mathrm{Tr}_{K_s/K}$ be the additive character of K_s (where typically $\psi'(\alpha) = \zeta_p^{\mathrm{Tr}(\alpha)}$). Denote Gauss sum as $g(\chi, \psi) := \sum_{\alpha \in K} \chi(\alpha)\psi(\alpha)$. Then,

$$(-1)^s g(\chi, \psi)^s = -g(\chi', \psi')$$

We need some auxiliary lemmas to prove this theorem. Further for monic polynomial $f(T) := T^n + \sum_{j=1}^{n} (-1)^j c_j T^{n-j}$, define $\lambda(f) := \psi(c_1)\chi(c_n)$.

Lemma 5.6. $\lambda(fg) = \lambda(f)\lambda(g)$ for all monic $f, g \in k[T]$.

Proof. Let $f(T) = T^n + \sum_{i=1}^{n} (-1)^i a_i T^{n-i}$ and $g(T) = T^m + \sum_{j=1}^{m} (-1)^j b_j T^{m-j}$. Note:

$$f(T)g(T) = T^{n+m} + \sum_{k=1}^{n+m} (-1)^k c_k T^{n+m-k}.$$

We are interested only in c_1 and c_{n+m}. Note $c_1 = a_1 + b_1$ and $c_{n+m} = b_m a_n$. Thus:

$$\lambda(fg) = \psi(a_1 + b_1)\chi(b_m a_n) = \psi(a_1)\psi(b_1)\chi(b_m)\chi(a_n) = \lambda(f)\lambda(g)$$

Lemma 5.7. Let $\alpha \in K_s$ and $f(T)$ be the monic irreducible polynomial for α over K. Then,

$$\lambda(f)^{s/d} = \chi'(\alpha)\psi'(\alpha), \quad d = \deg f$$

Proof. Note two things: if $K \subseteq K(\alpha) \subseteq K_s$ with $[K(\alpha) : K] = d$, then (a) $\mathrm{Tr}_{K_s/K}(\alpha) = (s/d)c_1$ and (b) $N_{K_s/K}(\alpha) = c_d^{s/d}$. Let us show them.

Recall $f(T) = (T - \alpha) \dots (T - \alpha^{q^{d-1}})$. Immediately $\mathrm{Tr}_{K(\alpha)/K} = c_1$ and $N_{K(\alpha)/K} = c_d$. Finally

$$\mathrm{Tr}_{K_s/K}(\alpha) = \mathrm{Tr}_{K(\alpha)/K}(\mathrm{Tr}_{K_s/K(\alpha)}(\alpha)) = \mathrm{Tr}_{K(\alpha)/K}([K_s : K(\alpha)]c_1) = \frac{s}{d}c_1$$

Similar logic can be applied to show $N_{K_s/K}(\alpha) = c_d^{s/d}$.

Now, recall that $\lambda(f) = \psi(c_1)\chi(c_d)$, thus

$$\lambda(f)^{s/d} = \psi(c_1)^{s/d}\chi(c_d)^{s/d} = \psi(c_1 s/d)\chi(c_d^{s/d}) = \psi(\mathrm{Tr}_{K_s/K}(\alpha))\chi(N_{K_s/K}(\alpha)) = \psi'(\alpha)\chi'(\alpha).$$

Lemma 5.8. $g(\chi', \psi') = \sum_{f \in \mathcal{P}_s} (\deg f)\lambda(f)^{s/\deg f}$ where f ranges over $\mathcal{P}_s$ — all monic irreducible polynomials f in $K[T]$ with $\deg f \mid s$.

Proof. By definition, $g(\chi', \psi') = \sum_{\alpha \in K_s} \chi'(\alpha)\psi'(\alpha)$. We partition the elements of K_s based on their minimal polynomials over K.

Let $\mathcal{P}_s$ be the set of all monic irreducible polynomials in $K[T]$ whose degree divides s. For any $\alpha \in K_s$, let $f_\alpha \in K[T]$ be the minimal polynomial of α. Since $K_s = \mathbb{F}_{q^s}$, we know that $\deg(f_\alpha)$ must divide s. Conversely, if $f \in \mathcal{P}_s$ with $d = \deg f$, then f splits completely in K_s and has exactly d distinct roots in K_s.

Thus, we can rewrite the summation by grouping roots of the same polynomial:

$$\sum_{\alpha \in K_s} \chi'(\alpha)\psi'(\alpha) = \sum_{f \in \mathcal{P}_s} \sum_{\alpha : f(\alpha)=0} \chi'(\alpha)\psi'(\alpha).$$

Let $f \in \mathcal{P}_s$ with $d = \deg f$. By **Lemma 2**, for any root α of f, the value of the character product is constant:

$$\chi'(\alpha)\psi'(\alpha) = \lambda(f)^{s/d}.$$

Since f has exactly d roots in K_s, the inner sum evaluates to:

$$\sum_{\alpha : f(\alpha)=0} \chi'(\alpha)\psi'(\alpha) = d \cdot \lambda(f)^{s/d}.$$

Summing over all $f \in \mathcal{P}_s$, we obtain:

$$g(\chi', \psi') = \sum_{f \in \mathcal{P}_s} (\deg f)\lambda(f)^{s/\deg f}.$$

Proof of Hasse-Davenport Relation.

Proposition 5.9. Let $\mathcal{M}(K)$ be all monic polynomials in $K[T]$. Then,

$$\sum_{f \in \mathcal{M}(K)} \lambda(f)T^{\deg f} = \prod_{f \in \mathcal{P}(K)} (1 - \lambda(f)T^{\deg f})^{-1}$$

For this identity not to look too random, it is analogous to the Euler product expansion $\sum_{n \geq 1} a(n)n^{-s} = \prod_p (1 - a(p)p^{-s})^{-1}$ for some multiplicative bounded function a.

Proof. Expand into geometric series:

$$\begin{aligned}
\text{RHS} &= \prod_{f \in \mathcal{P}(K)} (1 - \lambda(f)T^{\deg f})^{-1} \\
&= \prod_{f \in \mathcal{P}(K)} \sum_{n \geq 0} \lambda(f)^n T^{n \deg f} \\
&= \prod_{f \in \mathcal{P}(K)} \sum_{n \geq 0} \lambda(f^n) T^{\deg f^n} \\
&= \sum_{f \in \mathcal{M}(K)} \lambda(f)T^{\deg f}
\end{aligned}$$

The last equality more or less holds because each monomial in the former product is:

$$\prod_f \lambda(f^{n_f})T^{\sum_f n_f \deg f} = \lambda\left(\prod_f f^{n_f}\right) T^{\sum_f n_f \deg f}$$

Finally note that since $K[x]$ is a UFD, $\prod_f f^{n_f}$ corresponds to some monic polynomial from $\mathcal{M}(K)$ of degree $\sum_f n_f \deg f$, thus we get the desired result.

Now, for our λ, note that:

$$\sum_{f \in \mathcal{M}(K)} \lambda(f)T^{\deg f} = \sum_{n \geq 0} \sum_{\deg f = n} \lambda(f)T^{\deg f}$$

For $n = 0$, our sum is 1. For $n = 1$ we have:

$$\sum_{\deg f=1} \lambda(f) = \sum_{\alpha \in K} \lambda(T - \alpha) = \sum_{\alpha \in K} \chi(\alpha)\psi(\alpha) = g(\chi, \psi)$$

For $n > 1$, we have:

$$\sum_{\deg f=n} \lambda(f) = \sum_{c_i \in K} \lambda(x^n - c_1 x^{n-1} + \cdots + (-1)^n c_n) = q^{n-2} \sum_{c_1, c_s \in K} \chi(c_s)\psi(c_1) = 0$$

Now take logarithmic derivative of both sides and multiply by T:

$$\frac{g(\chi, \psi)T}{1 + g(\chi, \psi)T} = \sum_{f \in \mathcal{P}(K)} \frac{\lambda(f)(\deg f)T^{\deg f}}{1 - \lambda(f)T^{\deg f}}$$

Expand geometric series:

$$\sum_{n \geq 1} (-1)^{n-1} g(\chi, \psi)^n T^n = \sum_{f \in \mathcal{P}(K)} \sum_{n \geq 1} (\deg f)\lambda(f)^n T^{n \deg f}$$

Equating coefficient of T^s:

$$(-1)^{s-1} g(\chi, \psi)^s = \sum_{\deg f | s} (\deg f)\lambda(f)^{s/\deg f}$$

By Lemma above, RHS is $g(\chi', \psi')$.

References

[IR90] K. Ireland and M.I. Rosen. *A Classical Introduction to Modern Number Theory*. Graduate Texts in Mathematics. Springer, 1990. URL: https://books.google.com.ua/books?id=jhAXHuP2y04C.